VLTHR LAB

RESEARCH PLAN

A Comparative Analysis of How In-House and Outsourced Operations Affect a Company's Risk Profile

Applied research programme
Operational perspective • Strategic/business perspective • Evidence • Comparative analysis • Validation

Research activation condition

The research execution begins from Ground 0 after formal planning approval and confirmation of the 40% payment.

Data acquisition targets

Metric	Minimum target	Preferred target
Qualified company/professional contacts	250	300–500
Usable questionnaire responses	25	30–40
Planned minimum response rate	10%	Above 10% where achievable

Prepared by VLTHR LAB

Version 1.0 • Planning document

1. Research Plan Purpose

The plan defines the controlled process from research activation through data acquisition, analysis, development, validation and final reporting.

The research is positioned as applied business and operational research. It investigates how in-house, outsourced, hybrid and multi-sourced operating models redistribute, retain, transfer, introduce or amplify risk across operational and strategic/business dimensions.

2. Research Objective

Main objective: Develop a structured comparative understanding of how operating models affect a company's risk profile from operational and strategic/business perspectives.

Core research question: How does the choice between in-house and outsourced operations affect a company's risk profile from operational and strategic/business perspectives?

Supporting questions: What risks are retained, transferred, introduced or amplified? How does the operating model affect control, reliability, response, continuity, capability, dependency, flexibility and strategic exposure? Under what conditions do outcomes differ?

3. Research Positioning

The study should be presented as an applied investigation of a practical business problem, while remaining transparent about its academic status.

External positioning: “An applied research study examining how companies manage operational and strategic risk when activities are performed internally versus through external providers.”

The student/university identity remains visible for legitimacy and transparency. The primary reason for participation is the professional relevance of the business problem, not simply a request to help complete a thesis.

4. Reducing the “Student Seeking Data” Barrier

Academic legitimacy: identify the researcher, programme/institution, study title, supervisor where appropriate and research contact.

Business problem first: explain that the study examines how changing the ownership and delivery of operational activities changes control, dependency and risk.

Professional targeting: approach operations, management, procurement, supply chain, maintenance, engineering, HSE, risk, finance, IT, quality, business continuity and related professionals.

Low information burden: state completion time, avoid requests for confidential information, allow professional-level responses and explain data handling.

Professional value: offer an aggregated summary of research findings and the comparative risk framework after completion, without promising a particular result.

5. Research Population and Acquisition Targets

The acquisition campaign will target a minimum of **250 qualified company/professional email contacts**. The objective is not merely to collect addresses, but to build a relevant contact pool with organisational and respondent

context.

The minimum target is **25 usable responses**, equivalent to a planned 10% response rate from 250 contacts. A preferred target of 30–40 usable responses provides protection against later QA/QC exclusions.

6. Contact Data Structure

Company; industry; country/region; company size; contact name; job title; function; email; operating-model relevance; source/provenance; contact date; response status; follow-up status.

7. Sampling Strategy

Seek variation across predominantly in-house, predominantly outsourced, hybrid and multi-sourced operating models.

Where feasible, diversify by company size and industry. Potential sectors include oil & gas, energy, power, manufacturing, construction, logistics, telecommunications, technology, banking, healthcare, utilities, mining and FMCG. The final scope is to be frozen at Ground 0.

8. Research Architecture

Core causal chain: **Operating model** → **resource configuration** → **control distribution** → **dependency structure** → **risk exposure** → **risk event** → **consequence** → **recovery/response** → **residual risk.**

Operational dimensions: process control, quality, reliability, downtime, capacity, labour, skills, maintenance, information flow, response time, flexibility, continuity, supplier performance, technology dependency, safety and compliance.

Strategic/business dimensions: cost exposure, capital requirements, transaction costs, strategic dependency, vendor lock-in, core competence, scalability, flexibility, bargaining power, knowledge retention, IP, continuity, reputation and financial exposure.

9. Controlled Research Phases

The project uses 12 controlled phases, each with a validation gate. A failed gate returns the work to the required correction point before progression.

Phase	Stage	Primary output	Gate
0	Research Activation	Ground-0 baseline	G0
1	Problem Definition	Validated problem	G1
2	Research Architecture	Questions + methodology	G2
3	Knowledge Base	Literature framework	G3
4	Risk & Measurement	Risk model	G4
5	Data Acquisition	Evidence acquisition system	G5
6	Instrument Development	Final questionnaire/tools	G6
7	Data Collection	Primary + secondary dataset	G7
8	Data QA/QC	Validated dataset/evidence	G8

Phase	Stage	Primary output	Gate
9	Analysis & Modelling	Comparative findings/model	G9
10	Development	Research artefact/framework	G10
11	Validation & Synthesis	Validated conclusions	G11
12	Reporting	Final thesis	G12

10. Phase 0: Research Activation

Trigger: Research plan approved → 40% payment received → payment confirmed → research officially begins.

Activities: confirm scope, stakeholders and supervisor requirements; confirm research questions; establish repository, research log, evidence register, claim register, uncertainty register and validation register; confirm data protection, permission, consent and outreach requirements.

G0: scope, questions, permissions, data management, infrastructure and acquisition targets are sufficiently defined.

11. Phase 1: Ground-0 Problem Definition

Define the problem statement, operational and strategic perspectives, unit of analysis, operating-model categories, risk categories, boundaries, assumptions, unknowns, exclusions and terminology.

Output: Ground-0 Research Definition Pack. **G1:** the problem is precise enough to prevent materially different interpretations.

12. Phase 2: Research Architecture

Map research questions to variables, evidence requirements and analysis methods. Use the causal chain from operating model through resources, controls, dependencies, risk events, consequences and recovery.

G2: questions, variables, evidence and methods are aligned.

13. Phase 3: Knowledge Base

Acquire and review peer-reviewed research, books, theses, industry research, professional publications, standards/frameworks and company reports.

Investigate Transaction Cost Economics, Resource-Based View, Resource Dependence Theory, Agency Theory, Dynamic Capabilities, governance/relational theories and operational resilience, retaining only theories that materially explain the problem.

Each important claim should preserve source, year, context, population/sample, method, variable, finding, limitation, applicability and contradictory evidence.

G3: adequate coverage, theoretical foundation, research gap and connection to research questions.

14. Phase 4: Risk and Measurement Framework

Represent risk through exposure, probability, impact, detectability, controllability, transferability, recoverability and persistence.

Distinguish risk retained, transferred, introduced, reduced, amplified and shared.

G4: every major variable has a definition, measurement approach, evidence source and analytical purpose.

15. Phase 5: Data Acquisition Strategy

Literature: academic and professional evidence.

Primary company data: questionnaires, interviews and professional responses.

Secondary company data: annual reports, sustainability reports, operational disclosures, procurement/supplier information, incident information and other public evidence.

Case evidence: documented insourcing, outsourcing, hybrid operation, supplier failure, disruption, transition, concentration and recovery cases.

16. Phase 6: Questionnaire and Instrument Development

Develop the questionnaire from the validated research architecture rather than from generic questions.

Sections: respondent context; company context; operating model; activity characteristics; operational risk; strategic/business risk; dependency; control; capability; continuity/resilience; outsourcing/insourcing experience; risk outcomes; optional qualitative observations.

Pilot the instrument before mass distribution. Test comprehension, completion time, ambiguity, data structure and analysis compatibility.

G6: questionnaire/instruments pass pilot review.

17. Phase 7: Data Collection

Minimum campaign: 250 qualified contacts and 25 usable responses. **Preferred:** 300–500 contacts and 30–40 usable responses.

Use multiple outreach waves and run literature, case and secondary evidence acquisition in parallel so the study does not depend on survey data alone.

18. Phase 8: Data QA/QC and Evidence Validation

Screen responses for completeness, duplication, contradictions, implausible answers, respondent relevance, missing data, response patterns and insufficient engagement.

Classify evidence as accepted, conditionally usable or excluded, with reasons recorded.

G8: materially unresolved data quality issues are closed before core analysis.

19. Phase 9: Comparative Analysis

Compare in-house, outsourced, hybrid and multi-sourced models across control, cost exposure, dependency, capability, quality, reliability, response, continuity, flexibility, supplier risk, knowledge retention and strategic exposure.

The analysis must not assume that any operating model is universally superior.

20. Phase 10: Research Development

Convert validated findings into an Operating Model Risk Assessment Framework.

Input: activity, criticality, operating model, resources, supplier structure, capability requirements, dependencies and controls.

Logic: Operating Model → Dependency → Risk Mechanism → Exposure → Consequence → Response → Residual Risk.

Outputs: operational risk profile, strategic/business risk profile, risk transfer map, dependency map, control map and uncertainty assessment.

21. Phase 11: Validation and Synthesis

Validate against literature, primary data, documented cases and model logic. Perform sensitivity analysis and actively search for contradictory evidence.

G11: conclusions are accepted only where the evidence supports them and uncertainty is explicitly recorded.

22. Phase 12: Final Thesis Reporting

Final structure: Introduction; Knowledge Base; Methods; Analysis/Implementation; Results; Discussion; References; Appendices.

Discussion should address credibility, reliability, limitations, methodological decisions, ethical considerations, applicability and uncertainty.

23. Research Data Architecture

Experience / operating graph

Company → Activity → Operating Model → Resource → Dependency → Control → Risk → Risk Event → Consequence → Response → Outcome

Evidence graph

Source → Evidence → Claim → Finding → Relationship → Validation

This structure preserves provenance and allows evidence, claims, operational mechanisms and business consequences to be traced through the research.

24. Research Quality Controls

Control	Purpose
DQS	Completeness, consistency, validity, duplication and provenance checks.
DREV	Source quality, relevance, methodological strength, applicability and contradiction review.
UNPF	Classify evidence as known, probable, uncertain, unknown or unsupported.
CRDS	Claim → risk mechanism → evidence → operational consequence → business consequence → validation.
PURF	Progress certainty from reality and context through uncertainty, decomposition, system representation, state estimation, risk inference and validation.

25. Indicative Timeline

Period	Activity
Week 0	Payment confirmation + Ground 0
Weeks 1–3	Problem, research architecture and knowledge base
Weeks 2–4	Risk and measurement framework
Weeks 3–4	Data acquisition preparation
Week 4	Questionnaire pilot
Weeks 5–8	250+ contact campaign
Weeks 5–9	Interviews and secondary evidence
Weeks 7–9	Data QA/QC
Weeks 8–11	Comparative analysis
Weeks 10–13	Framework/model development
Weeks 12–14	Validation
Weeks 14–20	Results, discussion and final thesis

26. Success Metrics

• **Acquisition:** 250+ qualified contacts; preferably 300–500.

• **Response:** 25 minimum usable responses; preferably 30–40.

• **Evidence:** academic, industry, company and documented case evidence.

• **Quality:** usable-response rate, missing-data rate, duplicate rate, evidence validation rate and unresolved claims.

• **Output:** validated risk taxonomy, comparative operating-model framework, evidence-backed findings and validated research artefact.

27. Key Methodological Limitation

Twenty-five responses should be treated as a minimum exploratory target, not automatically as sufficient for broad statistical generalisation. The study therefore combines primary responses with literature, documented cases, secondary company evidence and professional evidence. If the final sample remains small, the limitation will be stated explicitly rather than implying population-level statistical certainty.

28. Final Research Gate Model

PLANNING → RESEARCH PLAN → APPROVAL → 40% PAYMENT → PAYMENT CONFIRMED → GROUND 0 → PROBLEM VALIDATION → RESEARCH ARCHITECTURE → KNOWLEDGE BASE → RISK + MEASUREMENT → DATA ACQUISITION → QUESTIONNAIRE/PILOT → DATA COLLECTION → DATA QA/QC → ANALYSIS → DEVELOPMENT → VALIDATION → RESULTS → DISCUSSION → FINAL THESIS → FINAL QA/SUBMISSION

29. Research Engagement Positioning

Recommended external statement:

“VLTHR LAB is conducting an applied research study into how organisations structure operational activities and how those choices affect operational and strategic risk. We are collecting evidence from professionals with direct experience in in-house, outsourced, hybrid or multi-sourced operations.”

The communication should remain transparent about the researcher's academic status while making the business problem, professional relevance, data protection and evidence contribution the primary reasons for participation.